

C.V. = Calorific value of coal.

3.80 Heat Flow

A power plant consists mainly of equipment for energy flow and transformation. The basic modes of heat transfer are as follows:

- (a) Conduction (b) Convection
(c) Radiation.

All the three forms of heat transfer are found in power plant engineering usually in combination with each other. The principal heat transfer calculations are associated with the following:

- (i) Radiation from fuel bed and luminous gases to boiler tubes and water walls.
- (ii) Radiation and conduction in heat insulators like refractories and pipe coverings.
- (iii) Conduction of heat through boiler, economiser and air preheater surfaces.
- (iv) Conduction of heat through condenser or heater tubes.
- (v) Heat convection from combustion region to more remote boiler tubes, economiser, tubes and air preheater surfaces.

Example 3.9. The overall thermal efficiency of a 40 MW turbo alternator is 30%. If the load factor of the power station is 50% and coal burnt has a calorific value of 6800 kcal, determine the following:

- (i) Coal consumption per kWh.
- (ii) Coal consumption per day.

Solution.

$$\text{Load factor} = \frac{\text{Average load}}{\text{Maximum demand}}$$

$$0.50 = \frac{\text{Average load}}{40 \times 1000}$$

assuming maximum demand equal to capacity of the power station.

$$\text{Average load} = 0.5 \times 40 \times 1000 = 20,000 \text{ kW}$$

$$\begin{aligned} \text{Energy produced per day} &= \text{Average load} \times 24 \\ &= 20,000 \times 24 = 48 \times 10^4 \text{ kWh.} \end{aligned}$$

Overall thermal efficiency

$$= \frac{860}{W \times 8600}$$

W = Coal consumption per kWh.

$$\therefore (1 \text{ kWh} = 860 \text{ kcal})$$

Assignment

$$0.3 = \frac{860}{W \times 8600}$$

$$\therefore W = \frac{860}{0.3 \times 8600} = 0.42 \text{ kg.} \quad \text{Ans.}$$

Coal required per day

$$= \text{Energy produced} \times W$$

$$= 48 \times 10^4 \times 0.42 \text{ kg.}$$

$$= \frac{48 \times 10^4 \times 0.42}{1000} \text{ tonnes}$$

$$= 201 \text{ tonnes.} \quad \text{Ans.}$$

Example 3.10. In a steam power station the coal consumption is 0.4 kg per kWh output. If the calorific value of coal is 6800 kcal per kg, boiler efficiency is 70% and mechanical and electrical efficiency of alternator is 90%. Prepare roughly the heat balance sheet for the power station.

Solution.

Let output = 1 kWh

$$\therefore \text{Heat input} = 0.4 \times 6100 = 2720 \text{ kcal.}$$

$$\text{Electrical energy input} = \frac{2720}{860} = 3.16 \text{ kWh}$$

Losses. (i) Boiler house loss

$$= 3.16 \times 0.3 = 0.948$$

(as boiler efficiency is 70%).

$$\text{Heat output of steam} = 3.16 - 0.948 = 2.212 \text{ kWh.}$$

$$\text{Input to alternator} = \frac{1}{0.90} = 1.11 \text{ kWh.}$$

(ii) Loss in alternator

$$= 1.11 - 1 = 0.11 \text{ kWh.}$$

(iii) Loss in turbine = 2.212 - 1.11 = 1.102 kWh.

This loss may be considered as rejected to cooling water.

Hence, output = 1 kWh.

Loss in boiler = 0.948 kWh.

Loss in alternator = 0.11 kWh.

Loss in turbine = 1.102 kWh.

Total = 1 + 0.948 + 0.11 + 1.102 = 3.61 kWh.

Example 3.11. *The daily output of a steam power station is 18×10^5 kWh. If the coal consumption is 700 tonnes per day, calculate the thermal efficiency of power station.*

Assume calorific value of coal 8500 kcal/kg.

Solution. Coal used = 700×1000 kg.

Energy input per day = $700 \times 1000 \times 8500 = 595 \times 10^7$ kcal

Energy output per day

$$= 18 \times 10^5 \text{ kWh}$$

$$= 18 \times 10^5 \times 860 \text{ kcal.}$$

(as 1 kWh = 860 kcal).

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} = \frac{18 \times 10^5 \times 860}{595 \times 10^7} = 26\%.$$

3.81 Cost of Steam Power Plant